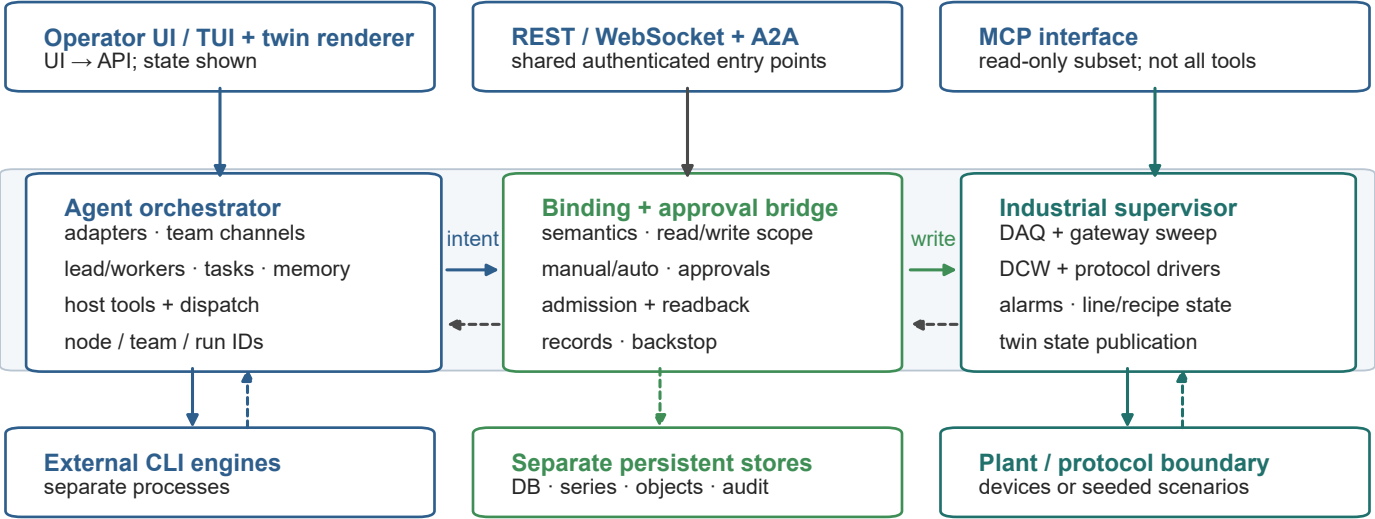
Operator UI / TUI + twin renderer
UI → API; state shown
REST / WebSocket + A2A
shared authenticated entry points
MCP interface
read-only subset; not all tools
Agent orchestrator
adapters · team channels
lead/workers · tasks · memory
host tools + dispatch
node / team / run IDs
intent
Binding + approval bridge
semantics · read/write scope
manual/auto · approvals
admission + readback
records · backstop
write
Industrial supervisor
DAQ + gateway sweep
DCW + protocol drivers
alarms · line/recipe state
twin state publication
External CLI engines
separate processes
Separate persistent stores
DB · series · objects · audit
Plant / protocol boundary
devices or seeded scenarios